

Zach Barbarich

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PROFESSIONAL SUMMARY

Full-stack developer and 11-year IT professional. Completed Carnegie Mellon full-stack certification and transitioned from IT operations to developing systems end-to-end: React/TypeScript frontends, Node.js microservices, PostgreSQL, Docker, cloud deployment.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, SQL, HTML5, CSS3

Frontend: React 18/19, Vite, Tailwind CSS, React Router, Recharts, responsive / mobile-first UI, dark mode, glassmorphism & interaction design

Backend: Node.js, Express, REST APIs, JWT, bcrypt, microservices, API gateways

Data: PostgreSQL (migrations, shared schemas), MongoDB

DevOps: Docker, Docker Compose, GitHub Actions, GitLab CI, DinD, Nginx, Caddy, Netlify, AWS (EC2, ECR, ECS, Route53, ACM), Oracle Cloud

Practices: Git, CI/CD, testing (Jest, Postman), Object Oriented Programming, reverse engineering / binary parsing, PyInstaller packaging

Systems / IT: Windows, Linux, macOS, Active Directory, VMware Horizon, network, system, & end-user support

PROJECTS

Andy Bot | *Field Service ERP | Solo | Live Production*

[Live](#) / [GitHub](#)

Stack: React 18, TypeScript, Vite, Tailwind, Node.js, Express, PostgreSQL 16, JWT, PDFKit, Docker Compose, Caddy, Oracle Cloud, GitHub Actions

- Designed and shipped a live single-tenant ERP covering customers → tickets → quotes/orders → deposits → invoices/payments → purchase orders, with printable PDFs and a KPI dashboard.
- Architected 7 Express microservices behind an API gateway (auth, customer, ticket, order, invoice, PDF) on a shared PostgreSQL 16 schema with 23 versioned migrations (000–022).
- Implemented JWT authentication with role-based access (`admin` / `tech`), gateway-enforced protected routes, global search, and Recharts revenue/A/R dashboarding.
- Deployed production on Oracle Cloud with Docker Compose + Caddy (automatic HTTPS) and GitHub Actions CI/CD (sequential builds for resource-constrained VPS).
- Built a polished React + TypeScript operator SPA (light/dark themes, glass UI, sticky sidebar, document/PDF workflows).

BidSpace | *Legacy .dat → Excel Converter | Solo | Workplace Production Tool*

[GitHub](#)

Stack: Python 3, openpyxl, tkinter, PyInstaller (Windows `.exe`)

- Reverse-engineered Bidtime's fixed ****2,501-byte**** binary record layout and built a production converter that outputs editable Excel workbooks (qty, SKU, manufacturer, cost/MSRP, unit/extended pricing).
- Handles A/B option pairing, continuation-line folding, combo/shipping sublines, vendor-specific parsing, and "NO BID" rows used in daily bidding workflows.
- Added an ED Data import normalizer so Northeastern-platform Excel imports stop failing on brand/SKU / "As Specified" mismatches.
- Packaged CLI + desktop GUI as portable Windows PyInstaller `.exe` builds used by staff (no install / no Python required); reduced large bid prep from ~2 hours to ~5 minutes.

React Portfolio | Solo | Live

[Live](#) / [GitHub](#)

Stack: React 19, Vite, Tailwind, Netlify Forms

- Built a UX-forward personal site that showcases production tools and full-stack systems with modern interaction design.
- Interactive canvas node-web background (hover / click / scroll), cursor-tracking hero mascot with parallax, 3D-tilt project cards, glass tech-stack carousel, and blur-fill media carousels.
- Contact via Netlify Forms; continuous deployment from GitHub to Netlify (migrated from AWS ECR/ECS for cost efficiency while retaining Docker/Nginx patterns in-repo).

Magnus Visibility | Production Availability ERP | CMU Capstone · Team of 3

[Live](#) / [GitLab](#)

Stack: React 18, Vite, Tailwind, Node/Express microservices, PostgreSQL, JWT, GitLab CI, Docker DinD, Nginx, AWS EC2

- Led front-end design and backend architecture for a manufacturing downtime/availability ERP (shift-linked logging, equipment tracking, MTD metrics, admin vs operator roles).
- Implemented JWT role-based access, API gateway routing/CORS, and a responsive light/dark React SPA for operators and admins.
- Shared ownership of GitLab CI/CD with Docker-in-Docker image builds and automated deployments to AWS EC2 + Nginx.

EDUCATION

Full Stack Development Certification | Carnegie Mellon University

Nov 2024 – Feb 2025

TalentSprint project-based bootcamp: React, Node.js, Express, PostgreSQL, DevOps / CI/CD.

Associates of Applied Science - Network Administration | Pennsylvania Highlands Community College

May 2013

EXPERIENCE

Accounts Receivable | Sportsman's

Nov 2025 – Present

- Own accounts receivable operations while solving workflow problems with code. Built and deployed BidSpace to eliminate manual legacy `.dat` bid transcription—reducing 2-hour bid prep to 5 minutes.
- Translate accounting and operations workflows into technical requirements; iterate with staff on Windows desktop tooling that improves how they work daily.

Laborer | Need a Hand Handy Man

Mar 2025 – Nov 2025

- Built full-stack systems and portfolio projects outside working hours while performing manual labor.

End User Services Analyst II (Help Desk L1) | American Red Cross

Jan 2018 – Aug 2024

- Supported 300+ users across multiple sites: ticket triage, Active Directory administration, VMware Horizon remote session management, and system troubleshooting.
- Monitored and escalated outages affecting critical services, biomedical systems, and production servers; collaborated with infrastructure teams to restore service quickly.

Solutions Specialist | Verizon Wireless

Sep, 2014 - Dec, 2017

- Delivered hardware and software troubleshooting across iOS and Android platforms; managed business accounts and recommended technical solutions based on customer infrastructure needs.